



UNSW
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MATH1231 Mathematics 1B – Algebra

Section 09 - Image, Kernel, and the Rank-Nullity Theorem

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Hidden similarities revealed ... again!



Exercise 1: Investigation. Consider the following three problems (two problems from algebra, and one from calculus).

- Given an $m \times n$ matrix A , solve the equation $A\vec{x} = \vec{0}$.
- Find in parametric vector form the equation of the plane which passes through the origin and is perpendicular to $(2, -1, 6)$.
- Solve the differential equation $y'' + 3y' + 2y = 0$.

Once again we have a situation in which a number of apparently different problems are really very similar.



The connection here is that all three problems involve solving an equation of the form $T(\vec{x}) = \vec{0}$, where T is a linear transformation.

The transformations are

- $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, where $T(\vec{x}) = A\vec{x}$;
- $T : \mathbb{R}^3 \rightarrow \mathbb{R}$, where $T(\vec{x}) = \vec{x} \cdot (2, -1, 6)$;
- $T : V \rightarrow V$, where V is the set of functions $y : \mathbb{R} \rightarrow \mathbb{R}$ such that all the derivatives $y', y'', \dots$ exist, and T is defined by
$$T(\mathbf{y}) = \mathbf{y}'' + 3\mathbf{y}' + 2\mathbf{y}.$$

Kernel

Definition (Kernel of a linear map)

Let $T : V \rightarrow W$ be a linear mapping. The **kernel of T** is the set of all $\vec{v}$ in V such that $T(\vec{v}) = \vec{0}$, that is,

$$\ker(T) = \{\vec{v} \in V \mid T(\vec{v}) = \vec{0}\}.$$

Definition (Kernel of a matrix)

Similarly, if A is an $m \times n$ matrix with entries in $\mathbb{R}$, then the **kernel of A** is

$$\ker(A) = \{\vec{x} \in \mathbb{R}^n \mid A\vec{x} = \vec{0}\}.$$

Comment. The kernel of a linear transformation (or a matrix) is also known as the **null space** of the transformation (or matrix).

Let's look at an example of the type of Problem (i) in the initial investigation.

Worked example 2. Consider the matrix

$$A = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 2 & 1 & -3 & 1 \\ 1 & -1 & 0 & 5 \end{pmatrix}$$

and consider the mapping $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by $T(\vec{x}) = A\vec{x}$.

a) Find $\ker(A)$.

b) Find $\ker(T)$.

Solution

a) $\ker(A)$ is the set of solutions of the equation $A\vec{x} = \vec{0}$. To solve $A\vec{x} = \vec{0}$, we reduce to echelon form,

$$(A | \vec{0}) \longleftrightarrow (U | \vec{0}) = \left(\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & -1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right),$$

and solve by back-substitution (exercise), giving

$$x_4 = \mu, \quad x_3 = \lambda, \quad x_2 = \lambda + 3\mu, \quad x_1 = \lambda - 2\mu.$$

Worked example 2, continued.

$$x_4 = \mu, \quad x_3 = \lambda, \quad x_2 = \lambda + 3\mu, \quad x_1 = \lambda - 2\mu.$$

Therefore

$$\begin{aligned}\ker(A) &= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} \lambda - 2\mu \\ \lambda + 3\mu \\ \lambda \\ \mu \end{pmatrix} \mid \lambda, \mu \in \mathbb{R} \right\} \\ &= \left\{ \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 3 \\ 0 \\ 1 \end{pmatrix} \mid \lambda, \mu \in \mathbb{R} \right\} \\ &= \operatorname{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 0 \\ 1 \end{pmatrix} \right\}\end{aligned}$$

Note that the kernel of A is a vector space (spans are always vector spaces).

b) $\ker(T)$? $\ker(T) = \ker(A)$, so same answer.

Example 3. (*This is Problem (ii) in the initial investigation.*)

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the linear map defined by

$$T(\vec{x}) = \vec{x} \cdot (2, -1, 6)$$

In other words, $T(\vec{x})$ is the dot product of $\vec{x}$ with $\vec{n} = (2, -1, 6)^T$.
We need to find $\ker(T)$.

Solution.

$$\ker(T) = \{ \vec{x} \in \mathbb{R}^3 \mid \vec{x} \cdot (2, -1, 6)^T = 0 \} = \{ \vec{x} \in \mathbb{R}^3 \mid 2x_1 - x_2 + 6x_3 = 0 \},$$

which is a plane through the origin (of course!).

We could solve the equation and express $\vec{x}$ in terms of two parameters, as we did in example 2; this would give the equation of the plane in parametric vector form, as requested.

 *Geometric interpretation:* The vector $\vec{n} = (2, -1, 6)^T$ is ... a vector normal to the plane $\ker(T)$.

Note that in this example, the kernel of T is a vector space.

Example 4. *(This is Problem (iii) in the initial investigation.)*

Let $T : V \rightarrow V$, where V is the set of functions $y : \mathbb{R} \rightarrow \mathbb{R}$ such that all the derivatives $y', y'', \dots$ exist, and T is defined by

$$T(\mathbf{y}) = \mathbf{y}'' + 3\mathbf{y}' + 2\mathbf{y}.$$

We need to find $\ker(T)$.

Solution.

We find the kernel of T by using standard calculus methods:

$$\begin{aligned}\ker(T) &= \{y \in V \mid \mathbf{y}'' + 3\mathbf{y}' + 2\mathbf{y} = \mathbf{0}\} \\ &= \{\lambda_1 \mathbf{e}^{-x} + \lambda_2 \mathbf{e}^{-2x} \mid \lambda_1, \lambda_2 \in \mathbb{R}\} \\ &= \text{span}\{\mathbf{e}^{-x}, \mathbf{e}^{-2x}\}\end{aligned}$$

Note that in this example, the kernel of T is a vector space.

Kernels are vector spaces

Notice that in each of these three examples, $\ker(T)$ is a vector space. This is not a coincidence! The kernel of a linear map is always a vector space:

Theorem (A kernel is a subspace)

Let $T : V \rightarrow W$ be a linear transformation.

Then $\ker(T)$ is a subspace of V , the domain of T .

Proof. Exercise! It's a standard use of the Subspace Theorem.

Comment.

- A similar result applies to the kernel of a matrix: if A is an $m \times n$ matrix with entries in $\mathbb{R}$, then $\ker(A)$ is a subspace of $\mathbb{R}^n$.
- Since $\ker(T)$ is a vector space, it makes sense to talk of its dimension.

Definition (nullity)

The **nullity** of a linear mapping T , or of a matrix A , is the dimension of its kernel.



A kernel is a *vector space*, and the nullity is a *number*, so they are completely different in nature: Do not mix them up!

Nullity of a linear map or a matrix: Examples

Example 5. Find the nullity of T for problems (i), (ii) and (iii) in the initial investigation.

Solution.

i) Here we have

$$\ker(T) = \ker(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 0 \\ 1 \end{pmatrix} \right\} ;$$

... these two vectors are linearly independent and so they form a basis for the kernel. Therefore $\text{nullity}(A) = 2$.

ii) In this case $\ker(T)$ is a plane and so ... its dimension is 2.

iii) Here $\{e^{-x}, e^{-2x}\}$ is a spanning set for $\ker(T)$. The two functions are ... linearly independent because they are not scalar multiples of each other; so they form a basis for $\ker(T)$, and once again $\text{nullity}(T) = 2$.

Nullity from echelon form

■ Let's look at Problem (i) more closely. By our previous working, we have

- $\text{nullity}(A) = 2$, because
- $\ker(A)$ has a basis consisting of 2 vectors, because
- the solution of $A\vec{x} = \vec{0}$ contains 2 parameters, because
- the echelon form of A has 2 non-leading columns.

Therefore, the nullity of a matrix can easily be read off its echelon form.

Fact (Nullity equals number of non-leading columns in REF)

The nullity of a matrix A is equal to the number of non-leading columns in an echelon form of A .

■ *What if there aren't any non-leading columns in an echelon form of A ?*

Firstly, this just means that $\text{nullity}(A) = 0$; on the other hand, it means that the equation $A\vec{x} = \vec{0}$ has a unique solution, and so the columns of A are linearly independent.

Fact (Nullity and linear independence)

The columns of a matrix A are linearly independent *if and only if* $\text{nullity}(A) = 0$.

Example 6. Consider the matrix A and its row echelon form U given below:

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 & -1 \end{pmatrix} \longleftrightarrow U = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

- a) Find the nullity of A .
- b) Find the kernel of A .

Solution.

- a) Simply by counting the non-leading columns of U , we have $\text{nullity}(A) = 3$.
- b) By solving the system (with zero right hand side) and manipulating the result, we obtain

$$\ker(A) = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Do the working to confirm this.



*Don't forget that the nullity of a matrix or of a linear transformation is just a **number**, while the kernel is a **vector space**.*

More examples

Example 7. *Two geometric examples in $\mathbb{R}^2$.*

- a) Let R be a rotation about the origin in $\mathbb{R}^2$ through an angle α . Find the nullity of R .
- b) Let $P : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the projection onto the x -axis: $P(x, y) = (x, 0)$. Find the nullity of P .

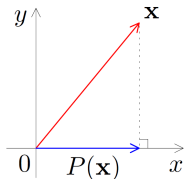
Solutions

- a) $R(\vec{x})$ is never the zero vector unless $\vec{x} = \vec{0}$. So the kernel of R is $\{\vec{0}\}$, and $\text{nullity}(R) = 0$.
- b) Clearly $P(x, y) = (0, 0)$ if and only if $x = 0$; thus

$$\ker(P) = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 \mid x = 0 \right\} = \left\{ y \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mid y \in \mathbb{R} \right\} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\},$$

hence $\text{nullity}(P) = 1$.

NB: The kernel is the y -axis. This can also be seen easily from a diagram.



More examples

Example 8. Define a linear mapping

$$T : \mathbb{P}_3 \rightarrow \mathbb{R} \quad \text{where} \quad T(p) = \int_{-1}^1 p(x) dx.$$

Find the kernel and the nullity of T .

Solution. We can write this as

$$T(a + bx + cx^2 + dx^3) = 2a + \frac{2}{3}c$$

so

$$\begin{aligned} \ker(T) &= \left\{ a + bx + cx^2 + dx^3 \mid 2a + \frac{2}{3}c = 0 \right\} \\ &= \left\{ a + bx - 3ax^2 + dx^3 \mid a, b, d \in \mathbb{R} \right\} \\ &= \left\{ a(1 - 3x^2) + bx + dx^3 \mid a, b, d \in \mathbb{R} \right\} \\ &= \text{span} \{ 1 - 3x^2, x, x^3 \} \end{aligned}$$

A basis for $\ker(T)$ is $\{ 1 - 3x^2, x, x^3 \}$, and therefore $\text{nullity}(T) = 3$.



Exercise 8: Another investigation which will reveal hidden connections.

Here are three further problems from algebra and calculus.

- Given an $m \times n$ matrix A and a vector $\vec{\mathbf{b}}$ in $\mathbb{R}^m$, does $A\vec{\mathbf{x}} = \vec{\mathbf{b}}$ have a solution?
- Let $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ be linearly independent vectors in $\mathbb{R}^3$. Given a vector $\vec{\mathbf{b}}$ in $\mathbb{R}^3$, does $\vec{\mathbf{b}}$ lie on the plane through the origin spanned by $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$?
- Does the differential equation $y'' + 3y' + 2y = e^{4x}$ have a solution?



All of these problems can be written in the form,

“Does the equation $T(\vec{\mathbf{v}}) = \vec{\mathbf{b}}$ have a solution?” ,

where T is an appropriate linear transformation.

The transformations are

- $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, where $T(\vec{\mathbf{x}}) = A\vec{\mathbf{x}}$;
- $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, where $T\left(\begin{pmatrix} \lambda \\ \mu \end{pmatrix}\right) = \lambda\vec{\mathbf{u}} + \mu\vec{\mathbf{v}}$;
- $T : V \rightarrow V$, where V is the set of functions $y : \mathbb{R} \rightarrow \mathbb{R}$ such that all the derivatives $y', y'', \dots$ exist, and T is defined by $T(\mathbf{y}) = \mathbf{y}'' + 3\mathbf{y}' + 2\mathbf{y}$.

Definition (Image)

Let $T : V \rightarrow W$ be a linear mapping. The **image of T** is

$$\begin{aligned}\text{im}(T) &= \{ \vec{w} \in W \mid T(\vec{v}) = \vec{w} \text{ has a solution} \} \\ &= \{ T(\vec{v}) \mid \vec{v} \in V \}.\end{aligned}$$

Similarly, if A is an $m \times n$ matrix with entries in $\mathbb{R}$, the **image of A** is

$$\begin{aligned}\text{im}(A) &= \{ \vec{b} \in \mathbb{R}^m \mid A\vec{x} = \vec{b} \text{ has a solution} \} \\ &= \{ A\vec{x} \mid \vec{x} \in \mathbb{R}^n \}.\end{aligned}$$

Comment. The image of a linear transformation T is also known as the **range of T** . In fact, *the image is just the same concept as the range of a function in calculus!* Compare

- if $f : \mathbb{R} \rightarrow \mathbb{R}$ and we have $y = f(x) = x^2$, then
$$\text{range}(f) = \{ x^2 \mid x \in \mathbb{R} \} = [0, \infty);$$

- if $T : \mathbb{R} \rightarrow \mathbb{R}^2$ and we have $T(x) = \begin{pmatrix} 2x \\ 3x \end{pmatrix}$, then

$$\text{im}(T) = \left\{ \begin{pmatrix} 2x \\ 3x \end{pmatrix} \mid x \in \mathbb{R} \right\} = \text{span} \left\{ \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}.$$

Let's look at an example of the type of Problem (i) in the initial investigation.

Example 9. Find the image of the linear map T defined by $T(\vec{x}) = A\vec{x}$, where

$$A = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 2 & 1 & -3 & 1 \\ 1 & -1 & 0 & 5 \end{pmatrix}.$$

Solution. To determine whether or not $A\vec{x} = \vec{b}$ has a solution we reduce an augmented matrix,

$$(A|\vec{b}) \longleftrightarrow (U|\vec{y}) = \left(\begin{array}{cccc|c} 1 & 0 & -1 & 2 & b_1 \\ 0 & 1 & -1 & -3 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 + b_2 - 3b_1 \end{array} \right).$$

From the echelon form we find

$$\text{im}(T) = \text{im}(A) = \left\{ \vec{b} \in \mathbb{R}^3 \mid b_3 + b_2 - 3b_1 = 0 \right\},$$

a plane through the origin.

Example 9, revisited. Note that we have seen all this before! The calculations above are exactly the same as those we would have performed in order to answer the question, “find conditions on b_1, b_2, b_3 such that $\vec{\mathbf{b}}$ is in the span of the columns of A ”. So we have in general

$$\text{im}(A) = \text{span} \{ \text{columns of } A \} = \text{col}(A) ,$$

the column space of A , which gives the alternative answer

$$\text{im}(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$$

in the present case.

since

$$A = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 2 & 1 & -3 & 1 \\ 1 & -1 & 0 & 5 \end{pmatrix} \longleftrightarrow U = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix} .$$

Example 10.

(This is Problem (ii) in the investigation.)

Recall that T is defined by $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, where $T \left(\begin{pmatrix} \lambda \\ \mu \end{pmatrix} \right) = \lambda \vec{u} + \mu \vec{v}$;

We want to find $\text{im}(T)$.

Solution. Here we have

$$\text{im}(T) = \{ \lambda \vec{u} + \mu \vec{v} \mid \lambda, \mu \in \mathbb{R} \} = \text{span} \{ \vec{u}, \vec{v} \} ,$$

and since $\vec{u}, \vec{v}$ are independent, this is a plane through the origin. Note, as in the previous example, a close connection between image and span.

Example 11. (*This is Problem (iii) in the investigation.*)

Recall that $T : V \rightarrow V$, where V is the set of functions $y : \mathbb{R} \rightarrow \mathbb{R}$ such that all the derivatives $y', y'', \dots$ exist, and T is defined by

$$T(\mathbf{y}) = \mathbf{y}'' + 3\mathbf{y}' + 2\mathbf{y}.$$

We want to find $\text{im}(T)$.

Solution. For $T(y) = y'' + 3y' + 2y$ it is not so easy to determine precisely what the image is. (Partly, this is because T is a transformation on an infinite-dimensional vector space, and all our matrix expertise is useless.) However, the image includes e^{4x} since we know from calculus how to solve the differential equation

$$y'' + 3y' + 2y = e^{4x}.$$

We also know from calculus that $\text{im}(T)$ contains

- any function e^{kx} , including e^{-x} and e^{-2x} ;
- any polynomial;
- any function $a \cos \omega x + b \sin \omega x$;
- ... and more!

Challenge. Show that the image of T also contains $f(x) = \frac{1}{x^2 + 1}$.



Like the kernel, the image of a linear map is a vector space.

Theorem (An image is a subspace)

Let $T : V \rightarrow W$ be a linear transformation. Then $\text{im}(T)$ is a subspace of W , the codomain of T .

Proof. $\diamond$ The image is by definition a subset of W , which is a known vector space.

$\diamond$ Because T is linear we have $T(\vec{0}) = \vec{0}$; therefore the equation $T(\vec{v}) = \vec{0}$ has a solution, and $\text{im}(T)$ contains the zero vector.

$\diamond$ Let $\vec{w}_1$ and $\vec{w}_2$ be vectors in $\text{im}(T)$. Then $\vec{w}_1 = T(\vec{v}_1)$ and $\vec{w}_2 = T(\vec{v}_2)$ for some vectors $\vec{v}_1, \vec{v}_2$ in V . Now T preserves addition, so we have

$$\vec{w}_1 + \vec{w}_2 = T(\vec{v}_1) + T(\vec{v}_2) = T(\vec{v}_1 + \vec{v}_2);$$

since $\vec{v}_1 + \vec{v}_2$ is a vector in V we have $\vec{w}_1 + \vec{w}_2 \in \text{im}(T)$, which shows that $\text{im}(T)$ is closed under addition.

$\diamond$ Let $\vec{w}$ be in $\text{im}(T)$ and let λ be a scalar. Again we have $\vec{w} = T(\vec{v})$ for some $\vec{v}$ in V ; since T preserves scalar multiplication,

$$\lambda \vec{w} = \lambda T(\vec{v}) = T(\lambda \vec{v}).$$

But $\lambda \vec{v}$ is in V and hence $\lambda \vec{w}$ is in $\text{im}(T)$; therefore $\text{im}(T)$ is closed under scalar multiplication.

We have shown that $\text{im}(T)$ is a subset of W which contains the zero vector and is closed under addition and scalar multiplication; therefore $\text{im}(T)$ is a subspace of W .

Comments

- Again we have a similar result for matrices: if A is an $m \times n$ matrix with entries in $\mathbb{R}$, then $\text{im}(A)$ is a subspace of $\mathbb{R}^m$.
- Note carefully that the image of a linear transformation $T : V \rightarrow W$ is a subspace of W , the codomain of T , whereas the kernel is a subspace of V , the domain of T . For an $m \times n$ matrix, the image is a subspace of $\mathbb{R}^m$ and the kernel is a subspace of $\mathbb{R}^n$.
- Observe the correct use of language in the previous proof: a linear transformation *preserves* addition and scalar multiplication, while a subspace *is closed under* addition and scalar multiplication.
- The dimension of the image, like the dimension of the kernel, is given a particular name.

Definition (rank)

The **rank** of a linear mapping T , or of a matrix A , is the dimension of its image.

Example 12. Let $A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 & -1 \end{pmatrix}$; find the rank of A .

Solution. We give two methods.

Method 1. Let's determine the vectors $\vec{\mathbf{b}}$ for which $A\vec{\mathbf{x}} = \vec{\mathbf{b}}$ has a solution. Augmenting A with a right hand side and row-reducing,

$$(A|\vec{\mathbf{b}}) = \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 0 & 0 & b_1 \\ 0 & 0 & 1 & 1 & 1 & b_2 \\ 1 & 1 & 0 & -1 & -1 & b_3 \end{array} \right) \longleftrightarrow \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 0 & 0 & b_1 \\ 0 & 0 & 1 & 1 & 1 & b_2 \\ 0 & 0 & 0 & 0 & 0 & -b_1 + b_2 + b_3 \end{array} \right).$$

Therefore


$$\text{im}(A) = \left\{ \vec{\mathbf{b}} \in \mathbb{R}^3 \mid A\vec{\mathbf{x}} = \vec{\mathbf{b}} \text{ has a solution} \right\} = \left\{ \vec{\mathbf{b}} \in \mathbb{R}^3 \mid \dots b_1 + b_2 + b_3 = 0 \right\}.$$

This is a plane through the origin; a plane is 2-dimensional, so A has rank 2.

Method 2. Alternatively, we use the fact that $\text{im}(A) = \text{col}(A)$ to obtain

$$\begin{aligned} \text{im}(A) &= \{ A\vec{\mathbf{x}} \mid \vec{\mathbf{x}} \in \mathbb{R}^5 \} = \text{span} \{ \text{columns of } A \} \\ &= \text{span} \{ \text{columns 1 and 3 of } A \} = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right\}; \end{aligned}$$

again we see that $\text{rank}(A) = 2$. Observe that this method gives $\text{im}(A)$ in terms of a basis, while the first gives conditions which ensure that $\vec{\mathbf{b}}$ is in $\text{im}(A)$.

The Rank–Nullity Theorem for matrices

The second method in the previous example should make the following result clear.

Fact (Rank equals number of leading columns)

The rank of a matrix A is equal to the number of leading columns in an echelon form of A .

This leads easily to the following very important result.

Theorem (The Rank–Nullity Theorem for matrices)

For any matrix A ,

$$\text{rank}(A) + \text{nullity}(A) = \text{number of columns of } A.$$

Proof. Clearly, the number of leading columns, plus the number of non-leading columns, equals the total number of columns of the matrix.

The Rank–Nullity Theorem for linear maps

Problem

In order to write this result in terms of a linear transformation rather than a matrix, we need to understand what “number of columns” corresponds to if we don’t have a matrix. Note that in the transformation

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad \text{where} \quad T(\vec{x}) = A\vec{x} ,$$

the number of columns of A must be the same as the dimension of the **domain** of T , or the matrix multiplication $A\vec{x}$ would not be possible.

Theorem (**The Rank–Nullity Theorem, generalised version**)

If V is a finite–dimensional vector space and $T : V \rightarrow W$ is a linear transformation, then

$$\text{rank}(T) + \text{nullity}(T) = \dim(V) .$$

Proof. Difficult, and not examinable for MATH1231. See the printed notes.

Example 13. We check that the Rank–Nullity Theorem holds for some examples from previous pages.

(a) Recall that given

$$A = \begin{pmatrix} 1 & 0 & -1 & 2 \\ 2 & 1 & -3 & 1 \\ 1 & -1 & 0 & 5 \end{pmatrix},$$

we reduced to echelon form,

$$(A | \vec{0}) \longleftrightarrow (U | \vec{0}) = \left(\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 0 \\ 0 & 1 & -1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right),$$

We have established that

$$\ker(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 0 \\ 1 \end{pmatrix} \right\}, \quad \text{im}(A) = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\},$$

so

$$\text{rank}(A) + \text{nullity}(A) = 2 + 2 = 4,$$

which is equal to the **number of columns of A** , as predicted.



Example 13, continued.

We check that the Rank–Nullity Theorem holds for some examples from previous pages.

(b) For $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ where $T(\vec{x}) = \vec{x} \cdot (2, -1, 6)$, we noted that the kernel of T is a plane.

The image is


$$\text{im}(T) = \{ 2x_1 - x_2 + 6x_3 \mid x_1, x_2, x_3 \in \mathbb{R} \} = \mathbb{R} ,$$

so

$$\text{rank}(T) + \text{nullity}(T) = 1 + 2 = 3 = \dim(\mathbb{R}^3) .$$

(c) For $T : V \rightarrow V$, where V is the set of functions $y : \mathbb{R} \rightarrow \mathbb{R}$ such that all the derivatives $y', y'', \dots$ exist, and T is defined by

$$T(y) = y'' + 3y' + 2y,$$

the Rank–Nullity Theorem does not apply since the domain V is infinite-dimensional.

Example 13, continued.

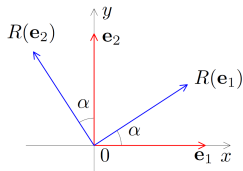
We check that the Rank–Nullity Theorem holds for some examples from previous pages.

(d) For the rotation example $R : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ we have

$$\ker(R) = \{ \vec{0} \} \quad \text{so} \quad \text{nullity}(R) = 0$$

$$\text{im}(R) = \mathbb{R}^2 \quad \text{so} \quad \text{rank}(R) = 2$$

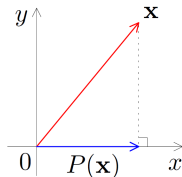
$$\text{and} \quad \dim(V) = 2 = \text{rank}(R) + \text{nullity}(R).$$



(e) And for the example of the projection of $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ onto the x-axis,

- the kernel is the y-axis,
- the image is the x-axis, so

$$\text{rank}(P) + \text{nullity}(P) = 1 + 1 = 2.$$



Example 13, continued.

We check that the Rank–Nullity Theorem holds for some examples from previous pages.

(f) Let $F : M_{3,3} \rightarrow M_{3,3}$ be given by $F(A) = A - A^T$. Then, as we proved earlier, F is a linear transformation. We have

$$\begin{aligned}\ker(F) &= \left\{ A \in M_{3,3} \mid A - A^T = \mathbf{0} \right\} \\ &= \{ \text{symmetric } 3 \times 3 \text{ matrices} \} \\ &= \text{span} \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\}\end{aligned}$$

and so F has nullity 6.

Example 13(f), continued.

From the Rank–Nullity Theorem, $\text{rank}(F) = \dim(M_{3,3}) - \text{nullity}(F) = 9 - 6 = 3$.
In fact,

$$\begin{aligned} F \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} &= \begin{pmatrix} 0 & b-d & c-g \\ d-b & 0 & f-h \\ g-c & h-f & 0 \end{pmatrix} \\ &= (b-d) \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + (c-g) \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} + (f-h) \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \end{aligned}$$

so the image of F is

$$\begin{aligned} \text{im}(F) &= \text{span} \left\{ \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \right\} \\ &= \{ \text{antisymmetric } 3 \times 3 \text{ matrices} \} \end{aligned}$$

confirming that $\text{rank}(F) = 3$.

Exercise. Find the kernel, image, nullity and rank for the linear transformation $F : M_{n,n} \rightarrow M_{n,n}$ defined by $F(A) = A - A^T$.

System of equations revisited with rank and nullity

Properties of linear equations $A\vec{x} = \vec{b}$ can be expressed in terms of rank and nullity.

Theorem (Rank, nullity and solutions of linear equation)

Consider the system $A\vec{x} = \vec{b}$.

- If $\text{rank}(A) \neq \text{rank}(A | \vec{b})$ then the system has no solutions.
- If $\text{rank}(A) = \text{rank}(A | \vec{b})$ then the system has solutions. In this case,
 - if $\text{nullity}(A) = 0$ then the solution is unique;
 - if $\text{nullity}(A) = k > 0$ then the general solution is

$$\vec{x} = \vec{x}_p + \lambda_1 \vec{v}_1 + \lambda_2 \vec{v}_2 + \cdots + \lambda_n \vec{v}_k,$$

where $\vec{x}_p$ is any solution of $A\vec{x} = \vec{b}$ and $\{ \vec{v}_1, \vec{v}_2, \dots, \vec{v}_k \}$ is a basis for $\ker(A)$.

Sketch of proof. $\diamond$ Row-reduce $(A | \vec{b})$ to echelon form $(U | \vec{y})$.

$\diamond$ We know from MATH1131 that the system has a solution if and only if the right hand side $\vec{y}$ is not a leading column. Since the rank of a matrix is equal to the number of leading columns in its echelon form, this is the same as saying that $\text{rank}(A) = \text{rank}(A | \vec{b})$. This proves the first statement and the beginning of the second.

$\diamond$ We have also seen in MATH1131 that the general solution can be written as $\vec{x} = \vec{x}_p + \vec{x}_h$, where $\vec{x}_h$ is the general solution of $A\vec{x} = \vec{0}$; but this means that $\vec{x}_h$ is an element of $\ker(A)$ and is therefore a linear combination of the basis vectors.

Examples

Example 14. Row-reducing an augmented matrix,

$$(A|\mathbf{b}) = \left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 1 \\ 1 & 1 & 5 & 2 & 0 \\ 2 & 7 & 0 & -1 & 3 \end{array} \right) \longleftrightarrow (U|\mathbf{y}) = \left(\begin{array}{cccc|c} 1 & 2 & 3 & 1 & 1 \\ 0 & -1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 \end{array} \right).$$

This system has no solution.

- **Reason** from MATH1131: there is an inconsistent row $(0 \ 0 \ 0 \ 0 \mid -2)$;
alternatively, the right hand side is a leading column.
- **Reason** from the theorem on the previous slide: $\text{rank}(A) = 2$ is different from $\text{rank}(A|\mathbf{b}) = 3$.

We see that the theorem gives us another way of stating what we learned in MATH1131, and helps us to understand the significance of the rank of a matrix.

Connection with Calculus: Maths is Maths is Maths

*A very important type of problem in calculus, with immense numbers of important applications in physics, engineering, mathematical finance and elsewhere, is that of solving **differential equations**. You will be learning some of the techniques for doing this in calculus lectures just about now.*

There are important connections between linear algebra and calculus. To illustrate this, we shall finish the chapter by confirming that that function from investigation (iii) on is linear, and recap some of its properties.

Example 15. Let V be the vector space of functions $y : \mathbb{R} \rightarrow \mathbb{R}$ such that all derivatives $y', y'', \dots$ exist, and define

$$T : V \rightarrow V \quad \text{where} \quad T(y) = \frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y .$$

First, recall that T is a linear map. Indeed,

- ◇ Then T is a function from one vector space to another (in this case, the same) vector space. Let y_1 and y_2 be functions in V ;
- ◇ then, using well-known properties of differentiation, we can show that T preserves addition and T preserves scalar multiplication.
- ◇ Thus T is linear.

Connection with Calculus: Maths is Maths is Maths

 We have seen that solving the homogeneous differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

is just the same as finding the kernel of T ;

 while asking whether a solution exists for the non-homogeneous differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = f(x) \quad (*)$$

is the same as asking whether $f(x)$ is in the image of T .

 Moreover, if the non-homogeneous equation has a solution, the general solution will take the form

$$y = y_p + \lambda_1 y_1 + \lambda_2 y_2 ,$$

where y_p is a particular solution of $(*)$ and $\{y_1, y_2\}$ is a basis for $\ker(T)$ and λ_1, λ_2 are arbitrary scalars: this is just like the solution of $A\vec{x} = \vec{b}$, see theorem *Rank, nullity and solutions of linear equations*.

Please remember that algebra and calculus are not really separate subjects: there are many important links between them. To increase your understanding of mathematics as a whole, you should always strive to be aware of these connections.